

Enhancing Keyword Extraction with KeyBERT: Evaluation, Optimization, and Transfer Learning

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Abstract—We investigate improvements to KeyBERT, a transformer-based keyword extraction model, through evaluation metric design, hyperparameter optimization, and post-processing. We demonstrate how relaxed evaluation metrics and fine-tuned transformer embeddings enhance topic relevance and clustering. Transfer learning further improves performance in cross-domain scenarios.

I. INTRODUCTION

Topic modeling stands as a foundational method within Natural Language Processing (NLP), enabling the discovery of hidden thematic structures across large collections of text. Traditional statistical models, such as Latent Dirichlet Allocation (LDA), have long served as the standard for extracting topics from unstructured data. However, recent progress in transformer-based models has opened new avenues for more nuanced and context-aware topic extraction. KeyBERT [1] utilizes BERT’s contextual embeddings to effectively identify keywords and keyphrases that best represent the underlying content of documents. This synergy between advanced keyword extraction and topic modeling—powered by transformer architectures—marks a significant evolution in the field, particularly as it enables transfer learning and adaptation to specialized domains. All the codes and results can be found at the project’s Github repository [2]

A. Research Objective

Our study addresses several key objectives:

- 1) Develop and Validate Comprehensive Evaluation Metrics to measure the coherence and relevance of extracted topics.
- 2) Systematically optimize KeyBERT’s hyperparameters for improved extraction performance.
- 3) Design, implement, and compare post-processing strategies to enhance the precision and relevance of extracted keywords.
- 4) Evaluate the capability of different transformer models to capture topic structure
- 5) Examine the transferability of topic modeling parameters across different domains.

B. Dataset

Our work utilized two primary datasets: *500N-KPCrowd-v1.1* and *SemEval2017 Task 10*.

- 1) *500N-KPCrowd-v1.1* is a broadcast news transcription dataset. This dataset consists of 500 English broadcast news stories from 10 diverse topics (art and culture, business, crime, fashion, health, US politics, world politics, science, sports, technology) with 50 documents per category.
- 2) *SemEval2017 Task 10* consists of 500 paragraphs selected from 500 ScienceDirect journal articles, evenly distributed among the domains of Computer Science, Material Sciences, and Physics. Each text has a number of keywords selected by one undergraduate student and an expert annotator.

These datasets were selected to ensure a comprehensive evaluation of keyword extraction models across both news and scientific publication domains.

II. METHODOLOGY

A. Baseline: Original KeyBERT

The original KeyBERT model extracts keywords by:

- 1) Encoding the document and candidate n-grams using a default transformer model.
- 2) Ranking candidates by cosine similarity to the document embedding.
- 3) Returning the top-N highest scoring candidates.

Most of our results contain a comparison between two techniques used by KeyBERT: Maximum Marginal Relevance (MMR) and MaxSum Distance. MMR selects keywords by balancing their relevance to the document with their dissimilarity to each other, promoting diversity in the output. In contrast, MaxSum Distance chooses keywords that maximize the overall semantic distance among them while maintaining relevance to the source text.

B. Evaluation Metrics

For each document, we assessed keyword extraction performance using two complementary evaluation strategies:

- **Strict Evaluation:** Predicted and reference keyphrases are matched exactly. A true positive is counted only when a predicted phrase is identical to the ground-truth. Precision, recall, and F1 score are computed using these exact matches, following standard definitions.
- **Relaxed Evaluation:** In this approach, both exact and partial matches are considered. An exact match occurs

when a predicted keyword is identical to a reference keyword. A partial match is recognized when a predicted keyword is a substring of a ground-truth or vice versa, provided it is not already counted as an exact match. This relaxed metric yields a more flexible assessment of extraction quality by rewarding meaningful overlaps in phrasing.

C. Post-Processing

To refine the set of candidate keywords returned by the model, we implemented a post-processing framework. This process incorporated two key components: named entity boosting and part-of-speech (POS) filtering. These enhancements were designed to align extracted keywords more closely with salient content in the source text while reducing noise from syntactically or semantically irrelevant phrases. We called advanced post-processing the combination of both components.

1) *Named Entity Boosting*: The first stage of post-processing targeted named entities—such as people, organizations, or locations—identified within the original document. Using a pre-trained dependency parser and named entity recognizer (NER) [3], each candidate keyword was compared against the list of extracted named entities. If a match was found, the candidate’s relevance score was boosted by a fixed additive factor (we used 0.2).

2) *Part-of-Speech Filtering*: We retained only those phrases composed entirely of tokens whose POS tags belonged to a predefined acceptable set, inspired by the feature-rich POS tagging approach of [4]. We considered the configuration: NOUN, PROPN, and ADJ, where NOUN is a noun like “dog”, “city”, or “happiness”, PROPN is a proper noun like “Paris”, “Einstein”, or “Microsoft”, and ADJ is an adjective like “red”, “large”, or “efficient”.

D. Hyperparameters

We then analyzed the impact of different parameters on the performance of KeyBERT. The evaluation pipeline systematically explored combinations of the following core hyperparameters:

- **N-gram Range** (1, 1), (2, 2), (3, 3): Defines the length of the candidate phrases. For example, (1, 1) restricts keywords to single words, while (3, 3) allows only trigrams.
- **Number of Candidates** (nr_candidates) [18, 19, 20, 21]: Controls the size of the initial candidate pool. A larger pool increases diversity but also computational cost.
- **MMR Diversity Parameter** (λ) [0.4, 0.5, 0.6]: Adjusts the balance between relevance and diversity in the selected keywords. Higher values emphasize diversity more strongly.

This resulted in 12 total configurations for MaxSum and 36 for MMR, forming a complete search space of 48 hyperparameter tuples. The number of top keywords extracted was fixed at 5.

E. Model Embeddings

To assess the performance of various sentence embedding models, we evaluated five pre-trained models. The evaluation focused on three core metrics: precision, recall, and F1 score.

F. Transfer Learning Strategy

To improve keyword extraction performance in low-resource settings, we implemented a hybrid transfer learning strategy using few-shot and zero-shot learning within the KeyBERT framework, which leverages Sentence-BERT for generating semantically meaningful embeddings [5].

Few-shot Pre-training: We selected a small subset of 20 labeled documents and their gold-standard keyphrases from the training set. This subset was used to warm-start the KeyBERT model through a supervised contrastive loss, helping it better align document and keyword embeddings in semantically meaningful ways. The model was fine-tuned for 3 epochs using a learning rate of $1e-3$.

Zero-shot Keyword Seeding: To guide the model toward domain-relevant concepts, we extracted the most frequent keywords from the labeled training set and used them as seed phrases for unsupervised domain adaptation. These seed keywords were fed into a custom `zeroshot_pre_train` routine, which adjusts the model’s keyword scoring heuristics via adaptive thresholding to reflect topic-specific salience.

Training Pair Generation: To train the contrastive objective, we constructed positive and negative pairs. Positive examples were formed by pairing each document with its ground-truth keyphrases. For negative sampling, we employed three strategies:

- Random n-grams extracted from the same document (intra-document noise),
- Keyphrases from unrelated documents (inter-document confusion),
- Low TF-IDF n-grams (semantically weak candidates).

This multi-strategy sampling ensured the model learned to differentiate relevant from non-relevant phrases, improving its generalization under both supervised and unsupervised scenarios.

III. RESULTS AND ANALYSIS

A. Evaluation metrics analysis

From our results, we can observe a stark difference between a strict evaluation metric and a relaxed one. In the 500N-KPCrowd dataset, we had a precision of 0.449 in the case of an exact match and 0.750 in the case of a partial match, representing a 67% increase. The full results are presented in Tables I and II. All the results in the following paragraphs are evaluated with the relaxed metric.

In addition to the quantitative evaluation metrics, we provide a qualitative visualization of topic groupings to assess the semantic coherence of the extracted keywords. Figure 4 presents

Metric Type	Precision	Recall	F1-Score
Strict	0.449	0.055	0.096
Relaxed	0.750	0.098	0.168

TABLE I
MODEL PERFORMANCE ON 500N-KPCROWD

Metric Type	Precision	Recall	F1-Score
Strict	0.200	0.059	0.088
Relaxed	0.865	0.285	0.415

TABLE II
MODEL PERFORMANCE ON SEMEVAL 2017

a 2D UMAP projection of document embeddings, where each point corresponds to a document and is color-coded by its dominant topic keyword.

The clusters show clear semantic separation between documents, confirming that KeyBERT can group similar documents under coherent topic labels. This qualitative insight supports the improved performance observed with partial matching metrics and highlights the model’s capacity to capture semantic similarity beyond exact surface form matches.

B. Post-processing analysis

We compared the performance of Named Entity Boosting alone, POS filtering alone, and advanced post-processing. In both datasets, the best results are reached when POS Filtering is applied. POS Filtering and advanced post-processing had the same scores, meaning that Entity Boost had no impact, which was confirmed by the fact that it had the same performance as using no post-processing. At the same time, POS Filtering improved the precision by only 1% in both datasets. The results are presented in Tables III-B and III-B

Post-processing	Precision	Recall	F1-Score
No Post-processing	0.750	0.098	0.168
Entity Boost	0.750	0.098	0.168
POS Filtering	0.759	0.086	0.149
Advanced	0.759	0.086	0.149

TABLE III
POST-PROCESSING 500N-KP

Post-processing	Precision	Recall	F1-Score
No Post-processing	0.865	0.285	0.415
Entity Boost	0.865	0.285	0.415
POS Filtering	0.872	0.246	0.368
Advanced	0.872	0.246	0.368

TABLE IV
POST-PROCESSING SEMEVAL

A similar analysis with comparable results was also done for the two specific techniques: MMR and MaxSum Distance. The tables are presented in Appendix B.

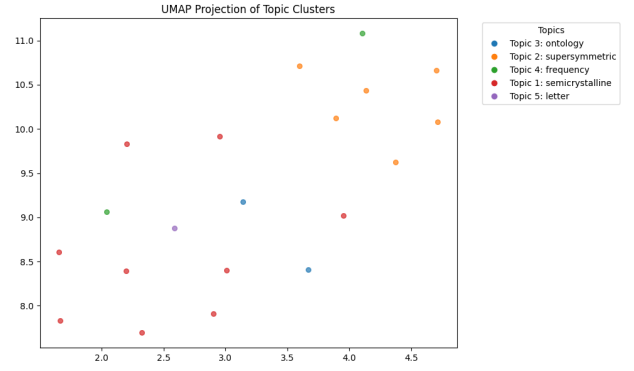


Fig. 1. UMAP projection of topic clusters generated using KeyBERT

C. Hyperparameters analysis

All configurations are evaluated with the same three metrics (precision, recall, and F1 score). To identify the best hyperparameters, we considered the configuration with the highest F1 score. The best hyperparameters are shown in Table III-C.

Dataset	Technique	Ngram	Candidates	Diversity	F1-Score
500N	MMR	(1, 2)	18	0.4	0.165
500N	MaxSum	(1, 2)	18		0.165
SemEval	MMR	(1, 1)	18	0.4	0.406
SemEval	MaxSum	(1, 1)	18		0.366

TABLE V
BEST HYPERPARAMETERS

For both datasets and both techniques, the n-gram hyperparameter is the one affecting the performance the most. We can see it in Figure 2. The other three figures with similar results are shown in Appendix

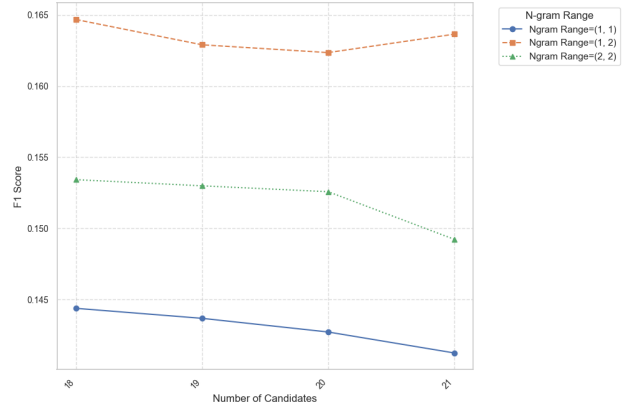


Fig. 2. MaxSum hyperparameters performance on 500N dataset

D. Model Embeddings

From our analysis, all-MiniLM-L12-v2 was the best-performing model on both datasets, slightly outperforming other models. The full results are presented in the tables III-D and III-D.

Model	Precision	Recall	F1-Score
paraphrase-MiniLM-L6-v2	0.763	0.100	0.170
paraphrase-MiniLM-L12-v2	0.748	0.097	0.166
all-MiniLM-L6-v2	0.750	0.098	0.168
all-MiniLM-L12-v2	0.767	0.100	0.171
paraphrase-albert-small-v2	0.760	0.100	0.170

TABLE VI
PERFORMANCE METRICS OF MODELS FOR 500N-KPCROWD

Model	Precision	Recall	F1-Score
paraphrase-MiniLM-L6-v2	0.832	0.277	0.401
paraphrase-MiniLM-L12-v2	0.849	0.280	0.407
all-MiniLM-L6-v2	0.865	0.285	0.415
all-MiniLM-L12-v2	0.873	0.290	0.420
paraphrase-albert-small-v2	0.840	0.278	0.403

TABLE VII
PERFORMANCE METRICS OF MODELS FOR SEMEVAL

E. Fine-tuning and Transfer Learning Results

To assess whether the base embedding model (all-MiniLM-L6-v2) could be improved through supervised contrastive fine-tuning, we trained the model on the SemEval2017 dataset. The original dataset was split into training, validation, and test sets. The fine-tuned model was then evaluated against the test split of SemEval2017 and compared with the baseline KeyBERT.

Model	Precision	Recall	F1 Score
Baseline KeyBERT	0.7184	0.0696	0.1233
Fine-tuned Model	0.7374	0.2196	0.3244

TABLE VIII
SEMEVAL2017: BASELINE VS. FINE-TUNED MODEL

To test generalization, we applied the fine-tuned model to the 500N-KPCrowd dataset without further training.

Model	Precision	Recall	F1 Score
Baseline KeyBERT	0.8467	0.0691	0.1233
Fine-tuned Model	0.8233	0.0739	0.1315

TABLE IX
TRANSFER LEARNING TO 500N-KPCROWD DATASET

IV. DISCUSSION

A. Evaluation Metrics

The better results computed through the relaxed evaluation metric can be explained by considering how sensitive the strict metrics are to small lexical or syntactic variations in the predicted keywords. Even when a prediction is semantically correct or closely related to a reference keyword, it is penalized if it does not match the reference verbatim. In practical applications, it can be useful to have the flexibility of using partial matching. These results confirm that partial matching can better reflect the model’s true capability.

B. Post-Processing

Post-processing did not improve our results as much as expected. In the case of Entity Boost, there was no improvement from baseline, which may mean that the keywords retained their original rank. If that is the case, it can be explained by a few scenarios:

- + 0.2 was too small a boost to make changes, e.g., keywords outside the top 5 get a boost, but their gap to the top 5 is larger
- some of the keywords in the top 5 got a boost, but they were already selected, so they did not change the result

In the case of the POS filtering, which removed any token not tagged as NOUN, PROPN, or ADJ, it increased the performance slightly by focusing on relevant words, but at the same time, it may have eliminated valid keywords. This balance may have caused the results we see.

C. Hyperparameter Analysis

In the hyperparameter analysis, n-gram was the parameter that affected the performance the most. This is explained by what n-gram does: determining which token spans are extracted as keyword candidates. The evaluation rewards exactly those n-gram lengths that the annotators chose when creating the dataset: if they focused more on one n-gram at the expense of another, choosing the right one drastically improved the performance. This theory is confirmed by the distribution in the two datasets: see Appendix A.

D. Fine-Tuning and Transfer Learning

Contrastive fine-tuning reshaped the embedding space to better capture domain-specific keyphrases: the model became better at detecting more relevant keyphrases (higher recall), while still filtering out most irrelevant ones (stable precision).

When transferred to the 500N-KPCrowd corpus, the model’s higher recall (even at a slight precision cost) reflects an increased willingness to propose candidate phrases in an unfamiliar domain. This behavior is coherent with contrastive fine-tuning: they become “risk-takers,” surfacing more potential matches at the expense of some false positives.

Overall, contrastive fine-tuning improved both in-domain performance and cross-domain robustness.

V. CONCLUSION

In this work, we demonstrated that careful selection of evaluation metrics, hyperparameter tuning, and targeted post-processing significantly enhance KeyBERT’s keyword extraction performance. Our experiments on 500N-KPCrowd and SemEval2017 confirm that relaxed matching and POS filtering yield higher F1 scores, while contrastive fine-tuning and zero-shot seeding improve recall in low-resource settings. We also showed that MiniLM-based embeddings consistently outperform alternatives across domains. These results underscore the importance of domain-aware model customization and provide actionable guidelines for transformer-based keyword extraction.

REFERENCES

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- [3] Kanerika, “Named entity recognition: A comprehensive guide to nlp’s key technology,” 2021. Accessed: 2025-07-11.
- [4] K. Toutanova, D. Klein, C. D. Manning, and Y. Singer, “Feature-rich part-of-speech tagging with a cyclic dependency network,” in *Proceedings of the 2003 Conference of the North American Chapter of the Association for Computational Linguistics on Human Language Technology-Volume 1*, pp. 173–180, Association for Computational Linguistics, 2003.
- [5] N. Reimers and I. Gurevych, “Sentence-bert: Sentence embeddings using siamese bert-networks,” in *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing*, 2019.

APPENDIX A GRAPHS DISTRIBUTIONS DATASETS

The following graphs show the length distribution of the ground truth keyphrases of the two datasets analyzed.

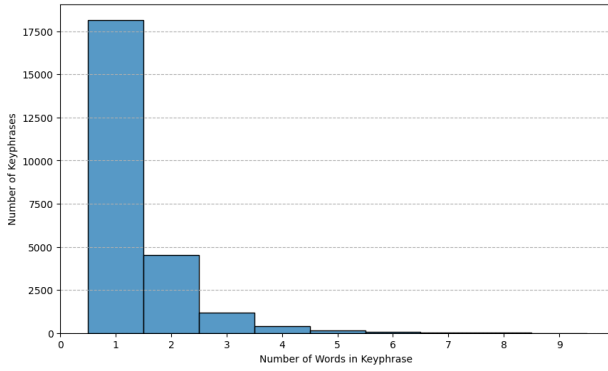


Fig. 3. Length Distribution of Ground Truth Keyphrases - 500N-KP

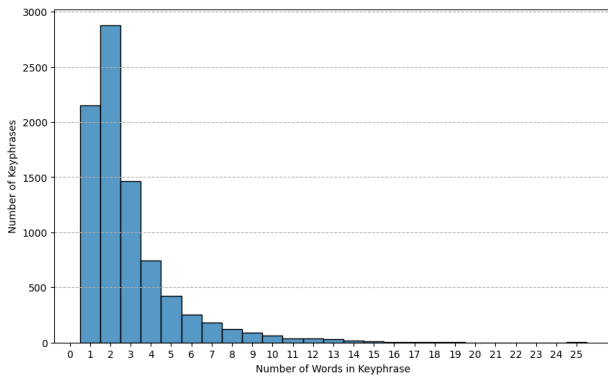


Fig. 4. Length Distribution of Ground Truth Keyphrases - SemEval

APPENDIX B MMR AND MAXSUM DISTANCE POST-PROCESSING

The following graphs compare the performance of using no post-processing with using it (POS filtering only).

Post-processing	Precision	Recall	F1-Score
No Post-processing	0.712	0.093	0.159
Post-processing	0.733	0.078	0.136

TABLE X
POST-PROCESSING MMR 500N-KP

Post-processing	Precision	Recall	F1-Score
No Post-processing	0.629	0.084	0.143
Post-processing	0.640	0.067	0.118

TABLE XI
POST-PROCESSING MAXSUM 500N-KP

Post-processing	Precision	Recall	F1-Score
No Post-processing	0.830	0.273	0.398
Post-processing	0.841	0.233	0.350

TABLE XII
POST-PROCESSING MMR SEMEVAL

Post-processing	Precision	Recall	F1-Score
No Post-processing	0.754	0.245	0.357
Post-processing	0.766	0.200	0.300

TABLE XIII
POST-PROCESSING MAXSUM SEMEVAL

APPENDIX C HYPERPARAMETERS ANALYSIS GRAPHS

Below are shown the performances of KeyBERT under different parameters.

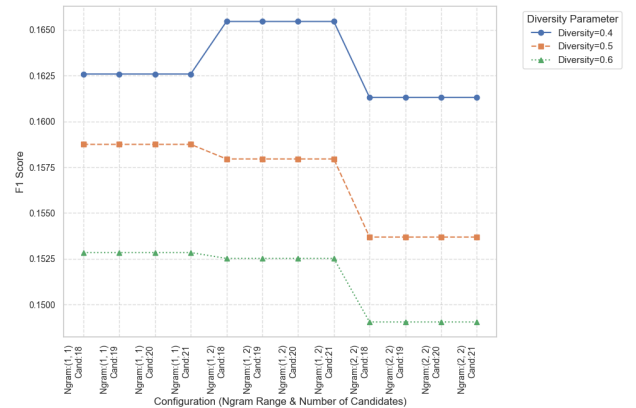


Fig. 5. MMR hyperparameters performance on 500N dataset

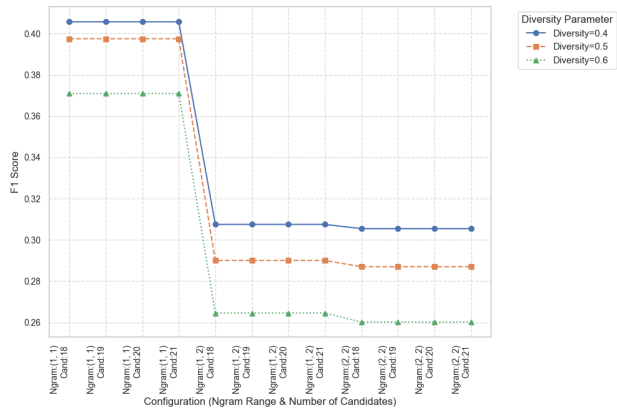


Fig. 6. MMR hyperparameters performance on SemEval dataset

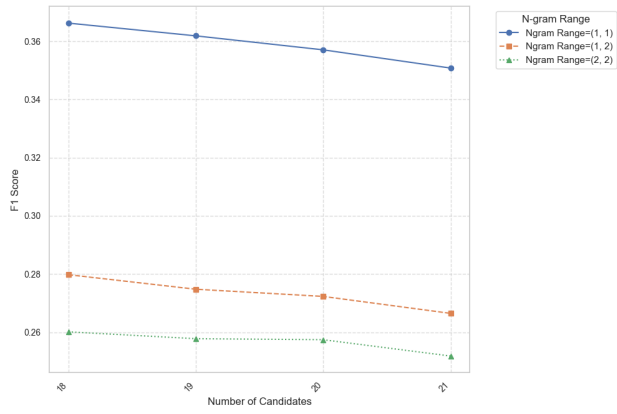


Fig. 7. MaxSum hyperparameters performance on SemEval dataset